

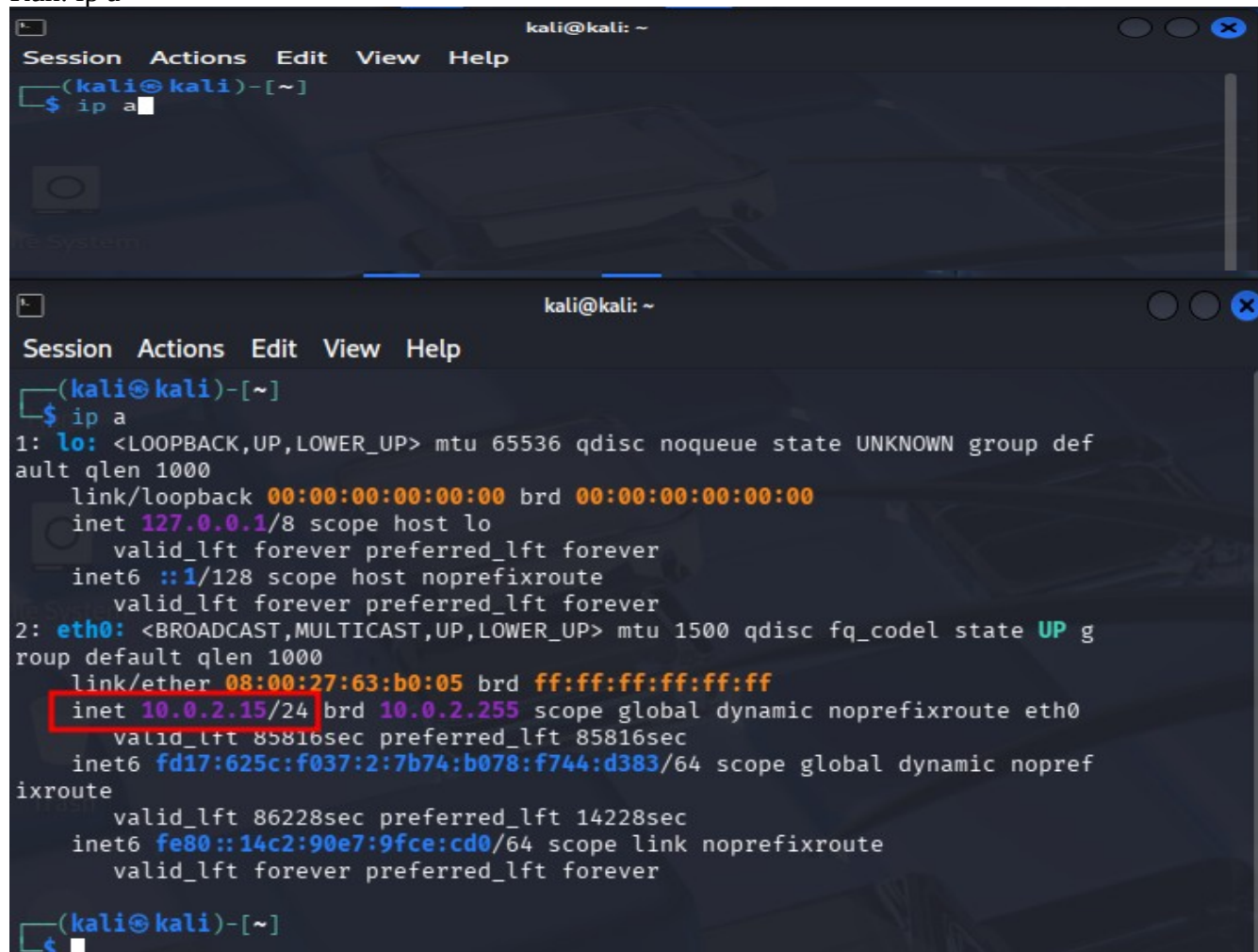
NETWORK SCANNING (RECONNAISSANCE)

This phase involved conducting network reconnaissance to identify active devices, open ports, and running services within the BSF-Nutrifeed test environment. Network scanning is a critical first step in cybersecurity assessment, as it reveals potential entry points that attackers could exploit.

Using tools such as Nmap within the Kali Linux virtual lab, controlled scans were performed to map the network structure and evaluate exposed services. The objective was to assess the attack surface of the BSF-Nutrifeed ecosystem and identify security gaps requiring remediation

1. Finding the local IP Address

Run: ip a



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ip a  
  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 85816sec preferred_lft 85816sec  
    inet6 fd17:625c:f037:2:7b74:b078:f744:d383/64 scope global dynamic nopref  
ixroute  
        valid_lft 86228sec preferred_lft 14228sec  
    inet6 fe80::14c2:90e7:9fce:cd0/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
  
(kali@kali)-[~]  
$
```

2. Discovering active device/s on my Network

Run nmap -sn 10.0.2.0/24

```

(kali@kali)-[~]
$ nmap -sn 10.0.2.15/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 18:16 EDT
Nmap scan report for 10.0.2.2
Host is up (0.00046s latency).
MAC Address: 52:55:0A:00:02:02 (Unknown)
Nmap scan report for 10.0.2.3
Host is up (0.00060s latency).
MAC Address: 52:55:0A:00:02:03 (Unknown)
Nmap scan report for 10.0.2.15
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.33 seconds

(kali@kali)-[~]
$

```

3. Scanning a specific IP for open ports

Run `nmap -sV 10.0.2.3`

```

(kali@kali)-[~]
$ nmap -sV 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 18:29 EDT
Nmap scan report for 10.0.2.3
Host is up (0.00084s latency).
Not shown: 999 filtered tcp ports (net-unreach)
PORT      STATE SERVICE VERSION
53/tcp    open  domain  dnsmasq 2.51
MAC Address: 52:55:0A:00:02:03 (Unknown)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 7.48 seconds

(kali@kali)-[~]
$

```

4. Performing deeper vulnerability scan

Run `nmap -A 10.0.2.3`

```

(kali@kali)-[~]
$ nmap -A 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 18:43 EDT
Nmap scan report for 10.0.2.3
Host is up (0.0017s latency).
Not shown: 999 filtered tcp ports (net-unreach)
PORT      STATE SERVICE VERSION
53/tcp    open  domain  dnsmasq 2.51
| dns-nsid:
|_ bind.version: dnsmasq-2.51
MAC Address: 52:55:0A:00:02:03 (Unknown)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|general purpose|bridge
Running (JUST GUESSING): AT&T embedded (92%), QEMU (90%), Oracle Virtualbox (90%), Slirp (90%)
OS CPE: cpe:/a:qemu:qemu cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp
Aggressive OS guesses: AT&T BGW210 voice gateway (92%), QEMU user mode network gateway (90%), Oracle Virtualbox Slirp NAT bridge (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE
HOP RTT      ADDRESS
1   1.71 ms  10.0.2.3

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 20.64 seconds

(kali@kali)-[~]
$

```

5. Scanning the entire Ports

Run nmap -p- 10.0.2.0/24

```
(kali@kali)-[~]
$ nmap -p- 10.0.2.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 18:53 EDT
Nmap scan report for 10.0.2.2
Host is up (0.00026s latency).
Not shown: 65522 closed tcp ports (reset)
PORT      STATE SERVICE
8000/tcp   open  http-alt
8065/tcp   open  unknown
8089/tcp   open  unknown
8191/tcp   open  limnerpressure
8194/tcp   open  sophos
9997/tcp   open  palace-6
34578/tcp  open  unknown
34907/tcp  open  unknown
39602/tcp  open  unknown
39911/tcp  open  unknown
45857/tcp  open  unknown
49614/tcp  open  unknown
54398/tcp  open  unknown
MAC Address: 52:55:0A:00:02:02 (Unknown)

Nmap scan report for 10.0.2.3
Host is up (0.00019s latency).
Not shown: 65534 filtered tcp ports (net-unreach)
PORT      STATE SERVICE
53/tcp     open  domain
MAC Address: 52:55:0A:00:02:03 (Unknown)

Nmap scan report for 10.0.2.15
Host is up (0.0000060s latency).
All 65535 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 65535 closed tcp ports (reset)

Nmap done: 256 IP addresses (3 hosts up) scanned in 12.26 seconds

(kali@kali)-[~]
$
```

6. Command Summary

- i) ip a = Local IP address
- ii) nmap -sn = Active devices on a Network
- iii) nmap -sV = Target IP (Open Ports)
- iv) nmap -A = Target IP (Deeper Vulnerability Scan)
- v) nmap -p- = Target IP (All IPs)

Findings

Network Scan Report — BSF-Nutrifeed Lab Environment

Tool Used: Nmap

Scan Type: Service & Version Detection

Date: (25/03/2026)

Target IP: 10.0.2.15/24

IP	Port	State	Service	Risk
10.0.2.2	8000/tcp	Open	http-alt	Web server vulnerabilities
“	8065/tcp	Open	unknown	Unidentified software may be outdated
“	8089/tcp	Open	“	“
“	8191/tcp	Open	limnerpressure	Rare service → may be misconfigured
“	8194/tcp	Open	sophos	If misconfigured, attackers may bypass security controls
“	9997/tcp	Open	palace-6	Legacy / uncommon service
“	34578/tcp	Open	unknown	Could be backdoor services
“	34907/tcp	Open	“	“
“	39602/tcp	Open	“	“
“	39911/tcp	Open	“	“
“	45857/tcp	Open	“	“
“	49614/tcp	Open	“	“
“	54398/tcp	Open	“	“
10.0.2.3	53/tcp	Open	domain	DNS spoofing

Potential Entry Points

Entry Point Type	Ports	Risk Level
Web Services	8000	High
DNS Service	53	High
Unknown Services	Multiple	High
Dynamic High Ports	49xxx+	Medium–High

Recommendations

To reduce the attack surface, it is recommended to close unnecessary ports, configure firewall protections, secure exposed services, enforce strong authentication, and implement continuous monitoring. These measures will significantly improve the security posture of the BSF-Nutrifeed ecosystem.

Update

After reconfiguring the virtual lab network to a Host-Only Adapter setup following the installation of Metasploitable2 VM which was newly added to my VM list, IP addresses were reassigned. A follow-up scan was conducted to reflect the updated BSF-Nutrifeed ecosystem addressing scheme.

Updated IP table

Windows 11 VM	Kali Linux VM	Wazuh VM	Metasploitable2
192.168.56.103	192.168.56.102	192.168.56.100	192.168.56.101

Updated Nmap scan screenshot

```
(kali㉿kali)-[~]
$ nmap -sn 192.168.56.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-26 06:47 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0018s latency).
MAC Address: 0A:00:27:00:00:00 (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.0027s latency).
MAC Address: 08:00:27:F0:E0:9A (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap scan report for 192.168.56.101
Host is up (0.0027s latency).
MAC Address: 08:00:27:0D:DC:1B (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap scan report for 192.168.56.103
Host is up (0.0026s latency).
MAC Address: 08:00:27:E2:9A:68 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap scan report for 192.168.56.102
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 28.21 seconds

(kali㉿kali)-[~]
$
```